INCH-POUND

MIL-M-70705B (AR)
29 July 1991
SUPERSEDING
MIL-M-70705A
1 February 1988

MILITARY SPECIFICATION

MICROWAVE CHASSIS ASSEMBLY: 9361371

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Microwave Chassis Assembly, 9361371, herein referred to as microwave chassis, which is a subassembly of the Receiver-Transmitter, RT-860A, a part of Radar Set AN/VPS-2A (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-D-50669 - Diode Switch Drive Assembly, 10548602
MIL-R-70560 - Radar Set, AN/VPS-2A, 9360797

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361371 - Microwave Chassis Assembly
9342815 - Klystron Power Supply Assembly
9360797 - Radar AN/VPS-2
10548164 - IF Amplifier Assembly
10548602 - Diode Switch Drive Assembly
10549065 - Connector
10549175 - Oscillator Module
10549202 - Transmit Gate Assembly
10556081 - Connector
10558238 - Waveguide, Coupler
11748124 - Radar System Tester
11834450 - Klystron Beam Power Supply
9155C2039 - Connector

SPECIAL PACKAGING INSTRUCTION

PS9361371 - Packaging of Microwave Chassis Assembly

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361371, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the microwave chassis shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.1 Subassemblies. The following subassembly is covered by the indicated specification:

2A1A8 Diode Switch Drive Assembly, 10548602, MIL-D-50669

3.3 Environmental service conditions.

3.3.1 Vibration. The microwave chassis shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The microwave chassis shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

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3.3.3 Altitude (Non-operating). The microwave chassis shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The microwave chassis shall meet the requirements of 3.4, 3.5.1.1, 3.5.1.2, 3.5.1.5, 3.5.2.1, and 3.5.2.5 when stabilized thermally at any temperature between -40 deg.F and +155 deg.F.

3.3.4.2 Non-operating (storage). The microwave chassis shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.F for a period of 24 hours and +160 deg.F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The microwave chassis shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the microwave chassis.

3.5 Performance. The microwave chassis shall meet the following requirements with the inputs described in Table I (Input characteristics).

3.5.1 Receive mode.

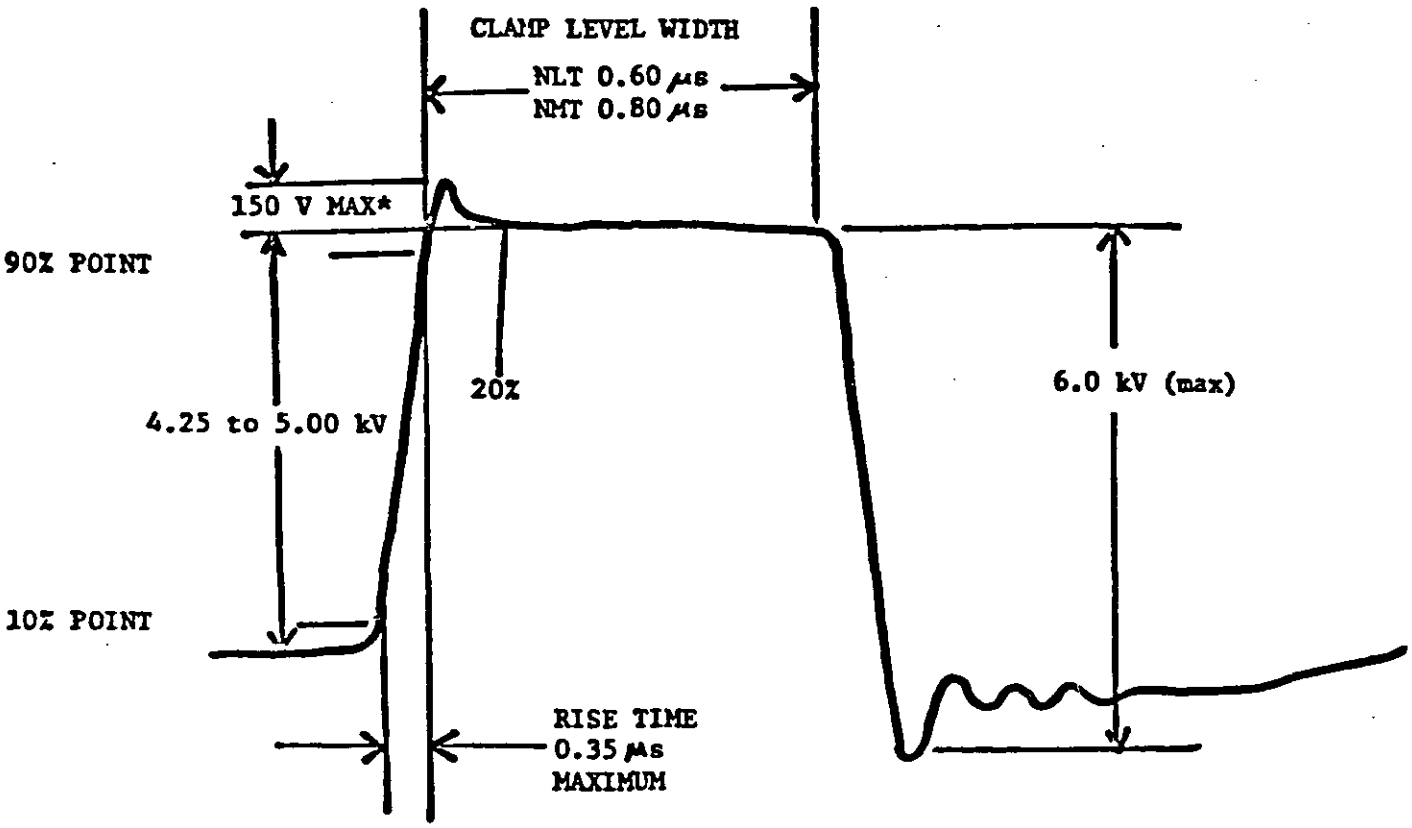
3.5.1.1 Overall gain. The overall gain of the microwave chassis, measured from the waveguide elbow (connected to HY1P2) to A5J2 with 0.0 Vdc at J1-A2, shall be 17 decibels (dB) minimum, 22 dB maximum, with an input level of -40 dBM.

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TABLE I. Input characteristics.

REQUIREMENTS	CHARACTERISTICS	CONNECTION
Input power	+28.0 ± 0.1 Vdc +28 Vdc return -33.0 ± 0.1 Vdc -33 Vdc return	J1-1, 3 J1-2, 4 J1-7 J1-8
AGC control voltage	0.0 to 6.0 Vdc	J1-A2
Diode switch driver trigger	PRF 19.978 KHz +1 Hz Pulse width 0.5 us Amplitude +18 V minimum into 50 ohm load with a SWR of <1.15 to 1	J2
Modulation signal*	Gated 60 MHz burst PRF 19.978 KHz Hz Width 280 ± 10 ns (50% of peak) Amplitude 1.0 V pk-pk, minimum into 50 ohm load with a SWR of 1.15 to 1	2A1A2J1
Klystron power amplifier filaments**	-9.0 to -9.5 KVdc (Ref. beam power supply) 7.9 to 9.5 mA 5.0 to 5.8 Vdc 2.0 to 2.8 amps dc at a reference level of -9.0 to -9.5 KVdc	A3P4-B
	Return	A3P4-C
Klystron power amplifier grid** (refer to Figure 1)	Dc level -13.25 to -14.00 KVdc Pulse amplitude +4.25 KV to +5.00 KV (grid bias cut off minimum 4.25 KV) Rise time (10-90% points) 0.35 us maximum *** Pulse width 0.60 us to 0.80 us PRF 19.978 KHz +1 Hz	A3P4-E
Klystron power amplifier collector	Ground	A3-E1

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* Returning to 50 V maximum in 20% of pulse width.

FIGURE 1. Klystron power amplifier grid bias pulse (from klystron power supply).

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TABLE I. Input characteristics (continued).

* Signal derived from the Transmit Gate Assembly 10549202.
** Signal derived from the Klystron Power Supply 9342815 and the Klystron Beam Power Supply 11834450.
*** Limits are over temperature and production variations. Room temperature limits are 0.67 to 0.77 us.

3.5.1.2 AGC action. The gain of the microwave chassis shall be reduced by a minimum of 20 dB when the dc voltage at J1-A2 is changed from 0.0 Vdc to +6.0 Vdc. The gain versus AGC voltage curve shall be monotonic.

3.5.1.3 Noise figure. The noise figure for the microwave chassis shall be 12 dB maximum on all six channels (11 dB maximum at room ambient).

3.5.1.4 Switch drive pulse. The switch drive pulse at A7J1 shall exhibit the following parameters:

Negative amplitude: 22 Vdc minimum
Positive reference: +0.7 Vdc minimum, +1.1 Vdc maximum
Width: (50% points) 2.48 us minimum, 2.62 us maximum

3.5.1.5 Diode switch isolation. When the dc level at A7J1 is changed from +1.0 to -33.0 Vdc, the insertion loss of the diode switch shall increase a minimum of 45 dB.

3.5.1.6 Stamo oscillator test port. The microwave chassis shall have a waveguide port that provides a stamo oscillator frequency output at a power level of not less than +3 dBM and greater than 20 dB isolation from local oscillator.

3.5.2 Transmit mode.

3.5.2.1 Average output power. The average output power, at PRF rate of 19.978 KHz + 1 Hz and duty unit of 0.0054 minimum, as measured at the waveguide elbow (connected to HY1P2) shall be 4.26 watts (W), minimum for all channels.

3.5.2.2 RF drive adjustment. The microwave chassis shall have an attenuator that provides a means of varying the RF drive to the klystron power amplifier A1A3. Minimum attenuation shall be 0.3 dB maximum and the attenuation range shall be 28 dB minimum.

3.5.2.3 RF power monitor. The microwave chassis shall have an indicator which when transmitting will provide a relative indication of the RF power being transmitted. The microwave chassis shall have a dc output that will provide a relative indication of RF power being transmitted to the BIT assembly 2A8. There shall be an adjustment such that at 5 $\pm$ 0.1 watts power output, the voltage at J4 shall be 40 $\pm$ 3 mVdc into a 10KΩ load. There shall be an adjustment such that at nominal power output the indicator will read mid-scale.

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3.5.2.4 Transmit pulse envelope. The transmit pulse envelope, when detected and observed on an oscilloscope, shall have a cosine squared (COS^2) characteristic shape and shall be equal to the modulation signal pulse width +10, -0 nanosecond (ns) at the 50 percent amplitude points (refer to Figure 2).

3.5.2.5 Transmit frequency. The microwave chassis shall be capable of operation on six discrete frequencies by selecting one of six oscillator plug-in modules. The frequency of the transmitted signal shall be 9205 MHz ± 10 MHz using channel 1 oscillator module. Using the channel 1 measurement as a reference, the channel 2 through 6 output frequency shall be as follows:

Channel 2	Reference +8 ± 2 MHz
Channel 3	Reference +16 ± 2 MHz
Channel 4	Reference +24 ± 2 MHz
Channel 5	Reference +32 ± 2 MHz
Channel 6	Reference +40 ± 2 MHz

3.5.2.6 Output spectrum. The power of the 1st and 2nd order upper and lower side lobes of the transmit pulse shall be as follows:

Side Lobe	Power Below Main Lobe
1st upper/lower	Not less than 26 dB
2nd upper/lower	Not less than 40 dB

3.5.3 System performance.

3.5.3.1 Subclutter visibility. The microwave chassis shall meet the subclutter visibility requirements of MIL-R-70560 when installed in Radar Set AN/VPS-2A.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to the detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

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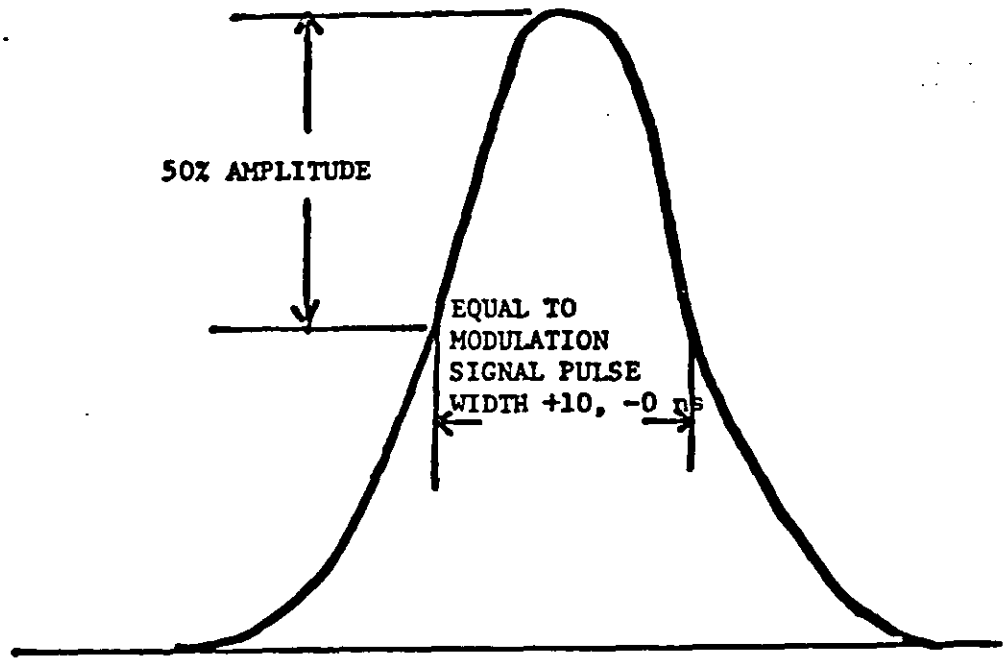


FIGURE 2. Transmit pulse envelope (detected RF).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg. F to + 90 deg. F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) microwave chassis for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II (Functional test) and Table III (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The

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tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One microwave chassis out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment should be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE IV. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. RF load	Bandwidth: 9150 to 9250 MHz minimum. Average power rating: 10 W minimum. VSWR: 1.0:1.1 maximum.
2. Crystal detector	Bandwidth: 9150 to 9250 MHz minimum. Average power rating: 10 mW minimum. VSWR: 1.0:1.8 maximum with 100 ohm load and 0.5 mA crystal current.
3. Directional coupler (2 required)	Bandwidth: 9150 to 9250 MHz minimum. Peak power rating: 2000 W minimum. Coupling: 40 dB ± 0.9 dB. Mainline VSWR: 1.0:1.1 maximum. Secondary line VSWR: 1.0:1.2 maximum.
4. Signal generator	Frequency range: 9150 to 9250 MHz minimum. Frequency stability: With temperature - less than 0.006%/ deg.C change in ambient temperature; with line voltage - less than 0.02% change for line voltage variation of ±10%. Residual FM: Less than 15 KHz peak. Output range: 0.224 V to 0.1 uV rms, into a 50 ohm load. Output calibration: 0 to -127 dBM. Source impedance: 50 ohm nominal.
5. Frequency meter	Type: Direct reading resonant cavity. Frequency range: 9150 to 9250 MHz minimum. Overall accuracy: 0.08%.
6. Thermistor mount	Frequency range: 9150 to 9250 MHz minimum. Maximum average input power: 10 mW minimum.
7. Power meter	Frequency range: 9150 to 9250 MHz minimum. Power range: 1 mW full scale. Accuracy: ±1% of full scale. Type: Thermistor sensing element.

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TABLE IV. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
8.	I-f amplifier	10548164, or equivalent.
9.	Noise figure meter	Type: Automatic readout with argon tube waveguide noise source and 60 MHz integral i-f amplifier. Scale: Not less than 20 dB full scale; 12 dB calibration point shall be not less than half scale.
10.	Waveguide noise source	Type: Argon-filled gas-discharge tube in standard waveguide house. Output connections: UG-39/U. Excess noise rating: 15.75 $\pm$ 0.5 dB.
11.	Oscilloscope	Vertical display: Dual trace, dc to 75 MHz (approximately 3 dB down) at 10 mV/cm minimum. Horizontal sweep: Calibrated time base 50 ns/cm to 1 s/cm minimum. Delayed sweep: 2 to 50 us, continuously variably, minimum.
12.	Power supply	Type: Laboratory supply. Minimum requirements: 0 to 32 Vdc. Load regulation: 0.02%. Load current capability: 15 A. Ripple and noise: 1 mV rms or 5 mVpeak-to-peak maximum from dc to 20 MHz.
13.	Power supply	Type: Laboratory supply, independent dual voltage source. Minimum requirements: 0 to 20 Vdc. Load regulation: 0.02%. Load current capability: 100 mA. Ripple and noise: 1 mV rms or 5 mV peak-to-peak maximum for dc to 20 MHz.
14.	Power supply	Type: Laboratory supply. Minimum requirements: 0 to 40 Vdc. Load regulation: 0.02%. Load current capability: 100 mA. Ripple and noise: 1 mV rms or 5 mV peak-to-peak maximum for dc to 20 MHz.

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TABLE IV. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
15.	RF voltmeter	Ranges: 3 to 300 mV in four ranges. Frequency range: 1 to 100 MHz minimum. Accuracy: ±3% of full scale.
16.	Digital voltmeter	Display: Four digit readout. Accuracy: 0.01%. Range: 100 Vdc.
17.	Pulse generator	Pulse amplitude: +18 V peak. Pulse width: 0.5 us. Rise time: 100 ns maximum.
18.	Test fixture	See Figure 3 and Table V.
19.	RF attenuator	Frequency range: Dc to 100 MHz. Attenuation: 0 to 30 DB in 1 DB steps.
20.	Transmit gate assembly	10549202, subassembly of Receiver-Transmitter, RT-860A/VPS-2A.
21.	Klystron power supply assembly	9342815, subassembly of Receiver-Transmitter, RT-860A/VPS-2A.
22.	Klystron beam power supply	11834450, subassembly of Receiver-Transmitter, RT-860A/VPS-2A.
23.	RF signal generator	Frequency: 59.936 MHz. Output level: 0.1 uV to 3.0 V rms into 50 ohms. Source impedance: 50 ohms nominal. Residual FM: Less than ±1 part in 10^6 or ±20 Hz peak, whichever is greater.
24.	Spectrum analyzer	Type: Laboratory. Frequency range: 10 MHz to 12 GHz. Sensitivity: -85 dBM. RF input attenuation: 0-60 dB. Spectrum width: 100 KHz - 2 GHz. Vertical display: Logarithmic 10 dB div.

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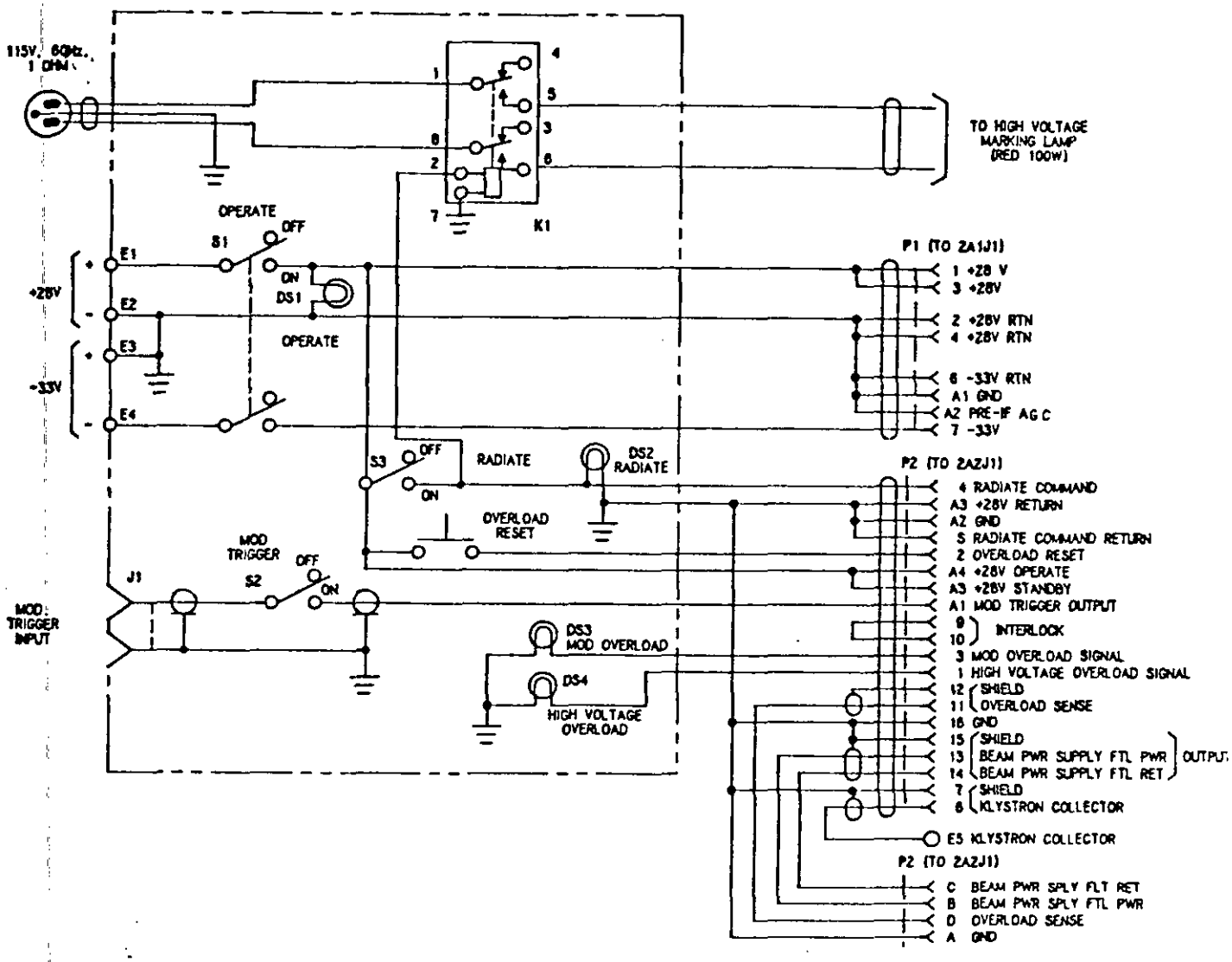


FIGURE 3. Test fixture schematic.

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TABLE IV. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
25.	Waveguide to "N" (female) adapter	Frequency range: 8.20 to 12.0 GHz. SWR:NMT 1.25 to 1.
26.	Waveguide, coupler	10558238
27.	Coaxial termination	Impedance: 50 ohms. VSWR: 1.15 to 1.
28.	Oscillator Modules Channels 1-6	10549175 - Oscillator Module
29.	Radar System Tester AN/VPM-2	11748124 or Equivalent
30.	Radar AN/VPS-2	9360797
31.	Resistor	10 kΩ, 1/2 W, 1%.

TABLE V. Test fixture parts list.

DESIGNATION	PART	DESCRIPTION
DS1 - DS4	Lamp 327	28 Vdc, 0.04 A
E1 - E4	Binding post	Superior, DF30, or equivalent
E5	Terminal lug	No. 8 size
J1	Connector	UG-625/U
K1	Relay	Magnecraft, W88CPX-16, or equivalent
MP1	Chassis	Bud, AC-404, or equivalent
P1	Connector	10556081
P2	Connector	10549065
P3	Connector	9155C2039
S1	Switch	DPDT, Cutler Hammer, 8823K5, or equivalent
S2	Switch	SPST, Cutler Hammer, 8282K14, or equivalent
S3	Switch	SPNO, Cutler Hammer, 8803K6, or equivalent
W1	Line cord	3 wire, 115 Vac
XDS1-XDS4	Lamp holder	Dialco, 95-0408-0931-241, or equivalent
XK1	Relay socket	Amphenol, 78RS-8, or equivalent

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The microwave chassis shall be attached to the vibration generator directly or with a fixture and securely held by its normal means of attachment. After vibration, the microwave chassis shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the microwave chassis shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (non-operating). This test shall be performed as specified in Procedures III, Method 500 of MIL-STD-810. The checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational in-chamber checkout shall be as specified in 4.7.2, 4.7.3.1, 4.7.3.2, 4.7.3.5, 4.7.3.7 and 4.7.3.11 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in

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3.3.4. The operational in-chamber checkout shall be as specified in 4.7.2, 4.7.3.1, 4.7.3.2, 4.7.3.5, 4.7.3.7, and 4.7.3.11 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. Connect the microwave chassis in the test setup shown in Figure 4, and adjust the test equipment as required in Table VI (Test equipment adjustments).

TABLE VI. Test equipment adjustments.

TEST EQUIPMENT	ADJUST FOR
Pulse generator (item 17, Table V)	PRF: 19,978 ± 200 Hz. Pulse width: 0.5 us. Pulse amplitude: +18 V (minimum).
Power supply (item 12, Table V)	+28 Vdc ± 0.1.
Power supply (item 14, Table V)	-33 Vdc ± 0.1.
Other items	As required by test procedure.

4.7.3.1 Overall gain.

a. Connect the signal generator (item 4, Figure 4) output to the waveguide elbow (connected to HY1P2) input/output port of the microwave chassis using the waveguide to N adapter (item 25, Figure 4). Set the output of the signal generator to -40 dBM.

b. Connect the RF voltmeter (item 15, Figure 4) to A5J2 mixer preamplifier.

c. Plug channel 4 crystal module into the X-band source (A1); tune the signal generator for a peak indication on the RF voltmeter. Set the AGC voltage control, R1, on the test fixture for 0.0 Vdc. The RF voltmeter shall indicate not less than -23 dBM and not more than -18 dBM. Verify compliance to 3.5.1.1.

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4.7.3.2 AGC action.

a. Disconnect the RF voltmeter and connect A5J2 to the attenuator (item 19, Figure 4), i-f amplifier (item 8, Figure 4) and oscilloscope.

b. Set 20 dB attenuation in the attenuator and establish a reference level on the oscilloscope, peaking up the signal generator tuning if necessary.

c. Advance the AGC voltage, using R1 on the test fixture, from 0.0 to +6.0 V at the same time reducing the attenuator to maintain reference level established in 4.7.3.2b. At 6.0 V AGC and 0 dB attenuation, the signal output shall not exceed the reference level to meet the requirements of 3.5.1.2.

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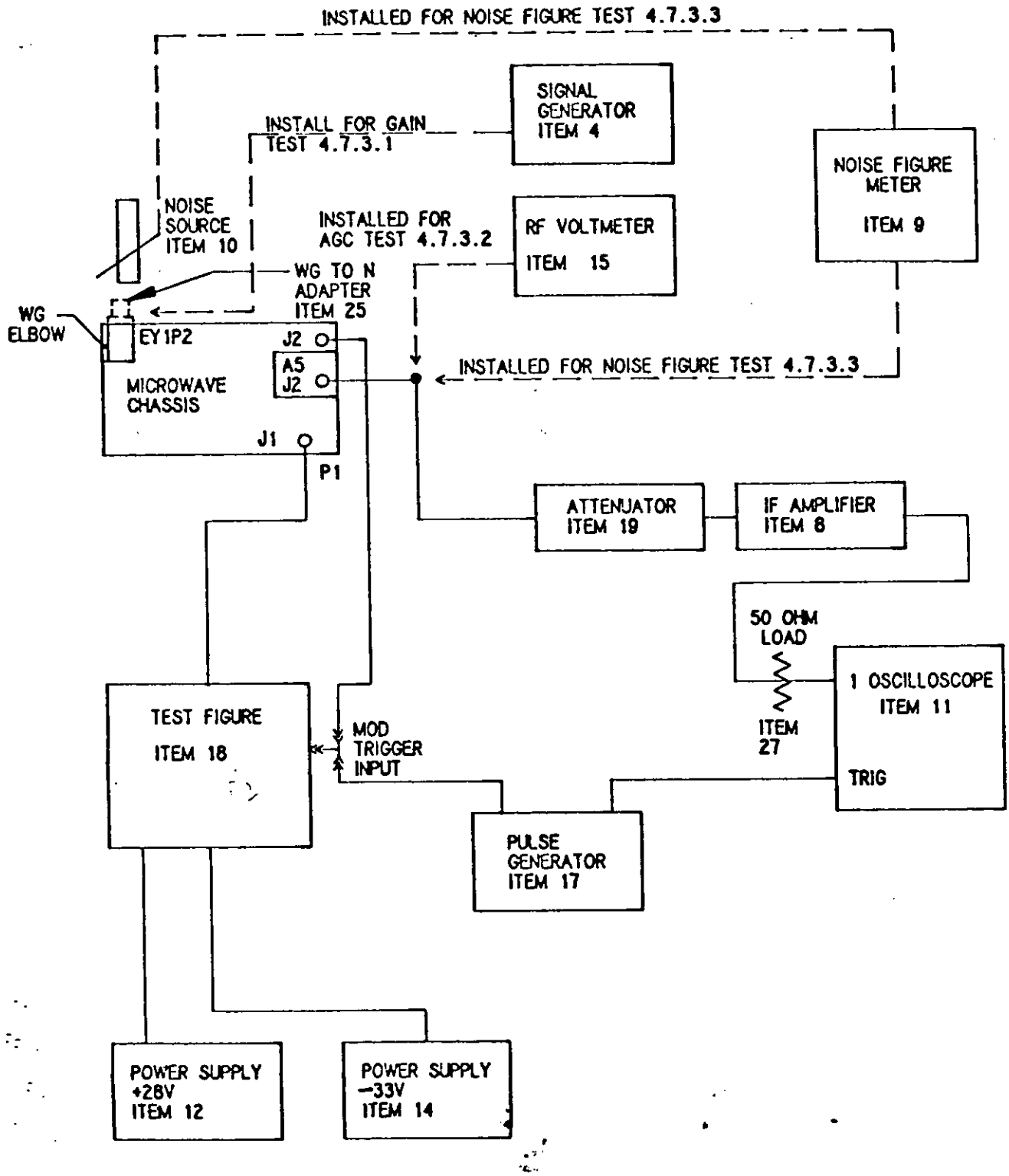


FIGURE 4. Test setup for receive operation.

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4.7.3.3 Noise figure.

a. Connect the waveguide noise source (item 10, Figure 4) to the waveguide elbow (connected to HY1P2) and connect the noise figure meter (item 9, Figure 4) to the source.

b. Connect a 50 ohm coaxial test cable from A5J2 to the i-f input on the noise figure meter.

c. Measure noise figure using the procedure given in the manufacturer's manual for the noise figure meter - reading shall be not more than 12 dB (11 dB maximum at room ambient).

d. Repeat c, installing each of the other five crystal modules, in turn, in the microwave assembly. The reading on each channel shall be not more than 12 dB (11 dB maximum at room ambient) to meet the requirement of 3.5.1.3. Disconnect the noise figure test equipment.

4.7.3.4 Switch drive pulse. Connect a test cable to 2A1A7J1 and 2P5, using a tee connector, and connect the other end of the cable to the oscilloscope channel 2 (item 11, Figure 4). The switch drive pulse parameters shall meet the requirements of

4.7.3.5 Diode switch isolation.

a. Connect the signal generator (item 4, Figure 4) to the waveguide elbow (connected to HY1P2) input port. Connect a test cable from preamp i-f output connector on the test fixture to channel 2 input on the oscilloscope with a 50 ohm load on the oscilloscope end.

b. Set the signal generator output level to -20 dBM and peak the frequency for maximum deflection on the oscilloscope.

c. Adjust the oscilloscope delayed sweep for a 5 us time base synchronous with pulse generator trigger (item 17, Figure 4).

d. With the display as shown in Figure 5, verify that the RF level during the gated off time is 45 dB or greater than the level during the on time. Verify compliance to 3.5.1.5.

4.7.3.6 Stamo oscillator test port. Connect the power meter (item 7, Table IV) with thermistor mount (item 6, Table IV), using waveguide coupler (item 26, Table IV) and waveguide to "N" adapter (item 25, Table IV), to the stamo oscillator test port of the microwave chassis. The power meter indication shall not be less than +3 dBM. Verify compliance to 3.5.1.6.

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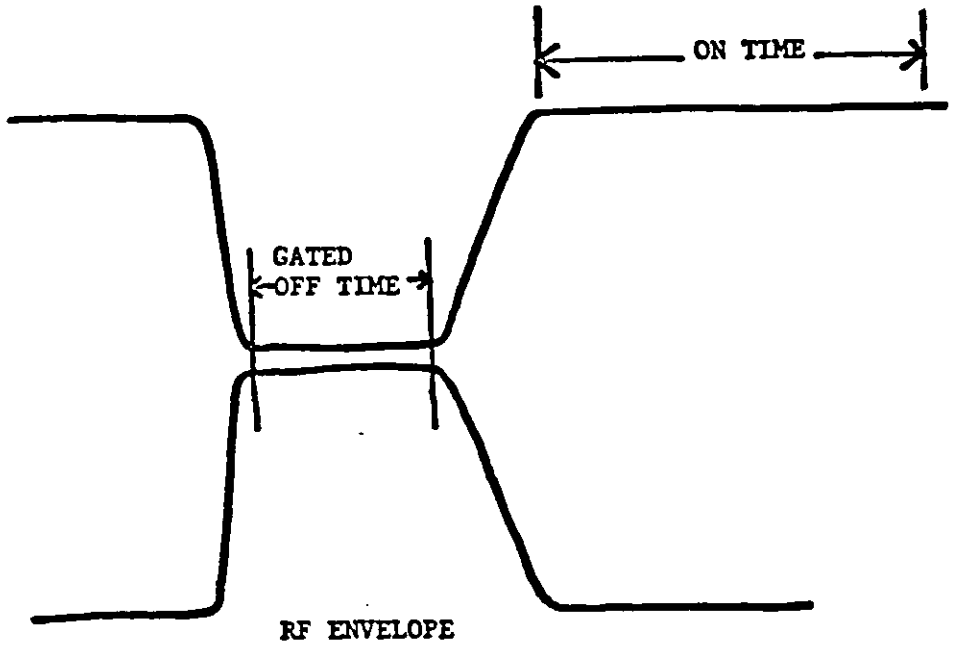


FIGURE 5. Diode switch isolation pattern.

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4.7.3.7 Average output power.

a. Connect the microwave chassis in the test setup as shown in Figure 6.

b. Turn on switch S3 of the test fixture and tune the klystron for maximum output at the operating frequency. The power meter (item 7, Figure 6) shall indicate not less than 0.426 mW. (Note: With 40 dB coupler, 0.426 mW at the coupled port indicates 1 KW peak pulse output at through port.)

c. Repeat b on each of the five channels to meet the requirements of 3.5.2.1.

4.7.3.8 RF drive adjustment. Remove the cap from AT1 and loosen lock screw. Turning the adjustment screw of AT1 shall vary the average power by more than 1.0 dB. Reset to nominal power level and tighten lock screw, replace cap. Verify compliance to 3.5.2.2.

4.7.3.9 RF power monitor. With a nominal 5 watt average power output (0.5 milliwatt on power meter), and a 10 Kohm load (Table V, item 31) connected between J4 center conductor and ground, adjust resistor 2A1R1 so that the RF power monitor reads mid-scale. Connect the DVM (item 16) positive lead to 2A1J4 center conductor and negative lead to chassis ground. Adjust 2A1R2 for a reading of 40±3 mVdc. Verify compliance to 3.5.2.3.

4.7.3.10 Transmit pulse envelope. With the microwave chassis in radiate condition, observe the detected RF envelope on channel 1 of the oscilloscope (item 11, Table IV). Check that the detected RF envelope (COS^2 pulse) is equal to modulation signal or NMT 10 ns wider at the 50 percent amplitude points, refer to Figure 2, to meet the requirements of 3.5.2.4.

4.7.3.11 Transmit frequency.

a. With test fixture switch S3 on, observe the detected RF pulse display, tune the cavity frequency meter (item 5, Figure 5) for a slight dip in pulse amplitude. At the point of maximum dip, read the frequency from the scale of the frequency meter. The reading shall meet the requirements of 3.5.2.5.

b. Repeat a with each of the other five crystal modules installed, tuning the klystron to each operating frequency to verify their compliance to 3.5.2.5.

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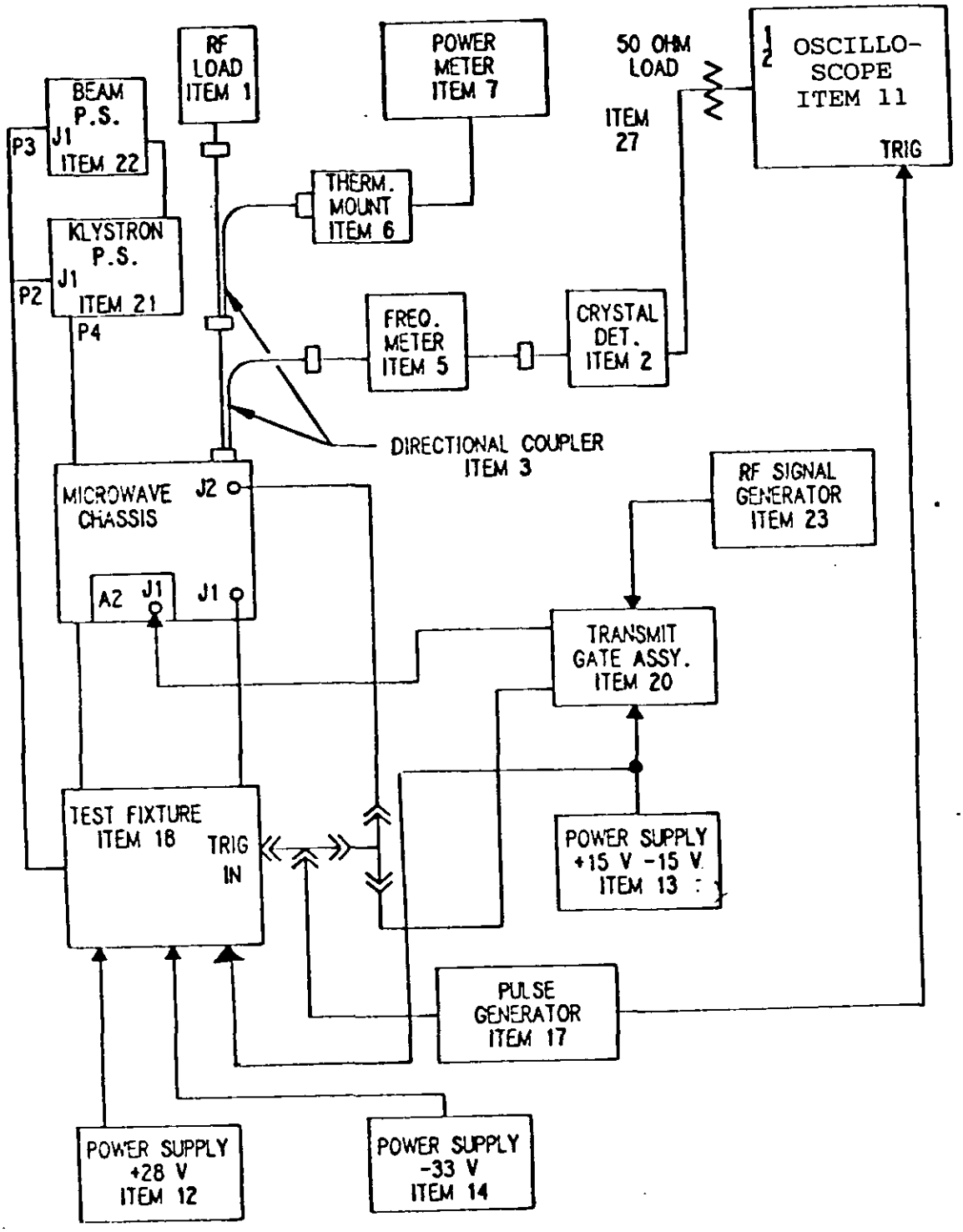


FIGURE 6. Test setup for transmit operation.

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4.7.3.12 Output spectrum.

a. Disconnect the crystal detector (item 2, Figure 6) from the frequency meter (item 5, Figure 6). Using an X-band test cable, connect the input of the spectrum analyzer (item 24, Table IV) to the frequency meter.

b. With test fixture switch S3 on, adjust the spectrum analyzer for a proper display. Verify the side lobes suppression compliance to requirements of 3.5.2.6.

4.7.3.13 Subclutter visibility. Verify compliance with 3.5.3.1 by installing the microwave chassis in Radar Set AN/VPS-2A, performing and meeting the subclutter visibility requirements of MIL-R-70560.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Special Packaging Instructions P9361371. Level of protection shall be as specified in the procurement document.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The microwave chassis is a subassembly of the Receiver-Transmitter, RT-860A, a part of Radar Set AN/VPS-2A. The receiver-transmitter provides sufficient RF power pulses to obtain usable echo return from targets of interest and converts these target signals to intermediate frequency for use in other units of the system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot of shipment or product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), May also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word listing).

Fire Control System
Product Improved Vulcan Air Defense System (PIVADS)
Radar Set AN/VPS-2A
Receiver-Transmitter RT-860A

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A754)

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